

ON THE DISCRIMINANT LOCUS OF A GENERIC PROJECTION

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ABSTRACT. The classical projective duality relates a projective variety X to its dual variety $X^\perp$, consisting of hyperplanes tangent to X . In this paper, we develop a relative version of this duality by studying the discriminant locus D of a generic projection of X . We show that the projective dual of the discriminant locus is isomorphic to a linear section of $X^\perp$. As an application, we prove that D is either a hypersurface or is empty, analogous to the Zariski–Nagata purity theorem for branched covers. In the case of projection to $\mathbb{P}^2$ and D is an irreducible curve, we study the monodromy group. In specific examples, we determine its singularities.

1. INTRODUCTION

Let X be a smooth projective variety over the $\mathbb{C}$. Given an embedding $X \subseteq \mathbb{P}^N$, a *Lefschetz pencil* on X is a choice of a general one-parameter family of hyperplane sections of X . There are finitely many singular fibers, and each has an ordinary nodal singularity. The singular fibers are in one-to-one correspondence with the intersection of a general line and the dual variety $X^\perp$ in dual space $(\mathbb{P}^N)^\vee$. Therefore, the number of singular fibers d^* equals the *class* of $X^\perp$, which is the degree of $X^\perp$ if it is a hypersurface, and zero otherwise. There is a class formula [Lam81, 3.5.3] that calculates d^* using the topological Euler characteristic of X , its hyperplane section, and the base locus of the pencil.

The motivation of this paper is to generalize this investigation to a *Lefschetz net*, and more broadly, a higher-dimensional linear subsystem of $|\mathcal{O}_{\mathbb{P}^N}(1)|$. Such a system arises from projectivization $\mathbb{P}V$ of a general $(k+1)$ dimensional subspace $V \subseteq H^0(\mathbb{P}^N, \mathcal{O}(1))$.

There are two natural families associated with the linear system. The first comes from observing that $\mathbb{P}V$ is naturally a subspace of the dual space $(\mathbb{P}^N)^\vee$, so the base change of the universal family of hyperplane sections of X over $(\mathbb{P}^N)^\vee$ to $\mathbb{P}V$ gives rise to the family

$$(1) \quad \mathcal{X}_V \rightarrow \mathbb{P}V,$$

whose fiber over $H \in \mathbb{P}V$ is the hyperplane section $X \cap H$.

Working dually, there is a generic projection

$$(2) \quad \pi : X \dashrightarrow \mathbb{P}V^\vee, \quad x \mapsto [H_0(x) : H_1(x) : \dots : H_k(x)],$$

where $H_0, \dots, H_k$ are $k+1$ general linear forms forming a basis of V . The base locus is the section by the linear subspace $\cap_{i=0}^k H_i$. Blow up the base locus, we can study the associated morphism instead.

These two families coincide only when $k = 1$, and they are precisely the Lefschetz fibration. However, when $k \geq 2$, their fibers differ: the first consists of hyperplane sections, while the second consists of higher codimension linear sections. We want to find the relationship between the two families, particularly where the fibers become singular.

1.1. Duality of discriminant locus.

Definition 1.1 (Discriminant locus). Let $\mathcal{Y} \rightarrow B$ be a flat family of projective varieties. Assume $\mathcal{Y}$ is smooth. Define the *discriminant locus* $D \subseteq B$ as the subspace consisting of points b where the fiber $\mathcal{Y}_b$ is singular.

The definition aligns with the classical discriminant [GKZ08] when the total space is smooth, but extends naturally to the singular cases. There, we define the discriminant to be the closure of the critical locus associated with the restriction $\mathcal{Y}^{sm} \rightarrow B$ to the smooth locus of $\mathcal{Y}$ (cf. Definition 3.1).

For the hyperplane sections family (1), the discriminant locus is precisely the linear section by $\mathbb{P}V$ of the dual variety $X^\perp \cap \mathbb{P}V$, which is reduced as long as $\mathbb{P}V$ is in general position, and is moreover irreducible if it has positive dimension. In this paper, we want to investigate the following questions:

Question 1.2. *How to characterize the discriminant locus D of the generic projection (2)?*

When $k = 1$, the discriminant locus consists of isolated points arising from critical values of Lefschetz fibration, and the degree is computed by the class formula [Lam81, 3.5.3]; When X is a surface, the generic projection $X \rightarrow \mathbb{P}^2$ is a branch cover and the discriminant locus D is an irreducible curve. In [CF11] and [CMT01], the authors showed that D has only nodes and cusps and computed their numbers based on the invariants of X . We would like to generalize this to higher dimensions.

We address these questions by constructing a natural birational model of the discriminant locus D based on projective duality.

Theorem 1.3. *Let $X \subseteq \mathbb{P}^N$ be an irreducible projective variety, and $\mathbb{P}V$ is a general k -dimensional linear subspace in the dual space $(\mathbb{P}^N)^\vee$. Let $D^\perp$ denote the dual variety of the discriminant locus of the generic projection (2), then there is an isomorphism*

$$D^\perp \cong \mathbb{P}V \cap X^\perp.$$

In particular, the discriminant loci of two families (1) and (2) are projectively dual to each other.

Note that when $k = N$, Theorem 1.3 is the classical projective duality (cf. Theorem 2.6). When π is not dominant, it is regular and is a generic immersion; the discriminant coincides with the image $\pi(X)$. In this case, a version of Theorem 1.3 is discussed in [Tev05, Theorem 1.21]. (Also see [Hol88, Proposition 6.1].) We are particularly interested in the case where the generic projection is dominant.

1.2. Purity of discriminant locus. Recall that the Zariski–Nagata purity theorem (cf. [Zar58, Nag58]) states that a finite morphism is either branched along a hypersurface (i.e., the branch locus has pure codimension one), or is étale with empty branch locus. We prove an analogue for the discriminant locus of generic projection.

From now on, $X \subseteq \mathbb{P}^N$ is an irreducible projective variety, and let

$$(3) \quad \pi : X \dashrightarrow \mathbb{P}^k$$

be a generic projection and assume it is dominant (equivalently, $\dim(X) \geq k$). For simplicity, we identify $\mathbb{P}V^\vee \cong \mathbb{P}^k$ once the generic projection is fixed.

Theorem 1.4. *Depending on the codimension of the dual variety $X^\perp \subseteq (\mathbb{P}^N)^\vee$, the discriminant locus D of the generic projection (3) satisfies one of the following:*

- (i) D is an irreducible hypersurface, if $\text{codim}(X^\perp) < k$;
- (ii) D is a finite union of hyperplanes, if $\text{codim}(X^\perp) = k$;
- (iii) D is empty, if $\text{codim}(X^\perp) > k$.

In particular, the discriminant locus of a dominate generic projection is either of pure codimension one or is empty.

We emphasize that, unlike in the branched cover case where the purity of the branch locus holds unconditionally, the purity of the discriminant locus in this setting relies on the genericity of the projection center. For example, projecting a quadric surface from a point lying on it yields a discriminant locus of codimension two (cf. Remark 3.13).

When X is smooth and $k \geq 2$, the generic cases are in (i). Indeed, smooth projective varieties with dual variety not a hypersurface are called *dual defective*, which are special (cf. Remark 2.9). The defect is defined to be $\text{def}(X) = \text{codim}(X^\perp) - 1$. In particular, case (iii) of Theorem 1.4 characterizes the dual defective varieties.

Corollary 1.5. *Suppose that X is smooth and is k -dual defective (i.e., $\text{codim}(X^\perp) > k$). Then after blowing up the base locus B , the map*

$$\tilde{\pi} : \text{Bl}_B X \rightarrow \mathbb{P}^k$$

is a smooth morphism.

As a special case, when $k = 1$, the discriminant locus is empty precisely when $X^\perp$ is not a hypersurface. Then X admits a Lefschetz fibration without singular fibers, e.g., X is the Segre threefold $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$ (cf. Example 5.3). Corollary 1.5 recovers a classical result of Ein [Ein86] that dual varieties for smooth curves and surfaces are hypersurfaces unless they are linear (see Proposition 5.2).

1.3. Degree of discriminant.

Use the polar locus characterization in terms of Chern classes, and that the image of the polar locus is the discriminant, we compute the degree of the discriminant curve.

Corollary 1.6. *The degree $\deg(D)$ of the discriminant locus of the generic projection (3) is the degree of the $(k - 1)$ -th polar locus of X . In particular, if X is*

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smooth, non-defective, then the discriminant of $X \dashrightarrow \mathbb{P}^2$ is an irreducible curve, whose degree is ...

1.4. Topology of the discriminant curve. As a consequence of Theorem 1.4, when X is a smooth projective variety and $X^\perp$ is a hypersurface, any generic projection $X \dashrightarrow \mathbb{P}^k$ with $k \geq 2$ has irreducible discriminant, and when $k = 2$, D is an irreducible curve. Suppose moreover the embedding $X \subseteq \mathbb{P}^N$ is sufficiently ample, then we can compute the degree of D , and the number of nodes and cusps of D (cf. Proposition 7.4). We computed this explicitly for curves and surfaces. See Proposition 6.9.

1.5. Braid groups and monodromy. For a choice of pencil on $\mathbb{P}^2$, a generic projection $X \dashrightarrow \mathbb{P}^2$ combined with this choice of pencil $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ gives a pencil of hyperplane sections of X as well as a pencil of pencils on hyperplane sections. Let $Y \hookrightarrow X$ be a general hyperplane section and $L \hookrightarrow \mathbb{P}^2$ be the corresponding general line of the chosen pencil, then we have the following commutative diagram

$$\begin{array}{ccccc} Y & \hookrightarrow & X & \xlongequal{\quad} & X \\ \downarrow \pi_Y & & \downarrow \pi_X & & \downarrow \pi \\ L & \hookrightarrow & \mathbb{P}^2 & \dashrightarrow^p & \mathbb{P}^1 \end{array}$$

where the monodromy group of π acts on Y and also on the pair $(L, L \cap D_{\pi_X})$ of L with the intersection of L with the discriminant locus of D_{π_X} . Write $\mathbb{O}_{X, \mathbb{P}^n}^{sm}$ for the smooth hyperplane sections of $X \hookrightarrow \mathbb{P}^n$, and let $\mathbb{O}_{D_{\pi_X}, L}^{sm}$ be the general hyperplane sections of D_{π_X} in L . It's clear that we have the following diagram of fundamental groups:

$$\begin{array}{ccc} \pi_1 \left(\mathbb{O}_{X, \mathbb{P}^n}^{sm}, Y \right) & & \\ \uparrow & \searrow \text{dashed} & \\ \pi_1 \left(\mathbb{P}^1 \setminus \text{crit}(\pi), * \right) & \twoheadrightarrow & \pi_1 \left(\mathbb{O}_{L, D_{\pi_X}}^{sm}, * \right) \end{array}$$

Theorem 1.7. *There exists a surjective group homomorphism $\pi_1 \left(\mathbb{O}_{X, \mathbb{P}^n}^{sm}, Y \right) \rightarrow \pi_1 \left(\mathbb{O}_{L, D_{\pi_X}}^{sm}, * \right)$ filling in the dashed arrow in the above diagram.*

Note that due to an exercise of [Voi07], when D_{π_X} is irreducible $\pi_1 \left(\mathbb{O}_{L, D_{\pi_X}}^{sm}, * \right)$ is isomorphic to the braid group of $\# \text{crit}(\pi)$ strands in $\mathbb{P}^1$. Therefore combined with theorem 1.4 we prove that all monodromy groups from Lefschetz pencils surject onto their corresponding braid groups.

Sketch of Proof. The proof of Theorem 1.3 relies on classical projective duality and standard linear algebra. The key step is to describe the critical locus of the projection π (cf. Proposition 3.5). We show that the natural map is birational between the critical locus and the discriminant locus.

For Theorem 1.4, in case (i), the irreducibility follows from Bertini's Theorem and the duality theorem 1.3. Cases (ii) and (iii) correspond to the intersection

$\mathbb{P}V \cap X^\perp$ consisting of finitely many points and being empty, respectively. The main difficulty lies in proving that D has codimension one in the case (i). We successively apply Theorem 1.3 to projections to a general hyperplane and reduce to the case of branch over, then we use the Zariski–Nagata theorem.

Remarks on Generalizations. One may ask whether Theorem 1.3 remains valid for special linear systems $\mathbb{P}V$ (specifically, that fails one of the conditions in Remark 3.7). This may require a version of projective duality in a certain limit sense (see [Zha23]). Another possible generalization is to consider a refined notion of discriminant of generic projection for a singular variety. Our definition 3.1 does *not* fully capture the information of the singularities of X (see Remark 5.4). We leave these questions to future work.

Finally, although we work over complex numbers, our results hold over any algebraically closed field of characteristic zero by the Lefschetz principle. In positive characteristic, the arguments may still apply provided the variety X is reflexive (i.e., double dual is isomorphic to X) and an appropriate form of Bertini’s theorem is available.

Structure of the paper. In Section 2, we discuss the linear algebra foundation that relates the two families (1) and (2) from linear algebra. We recall the classical projective duality for projective varieties. In Section 3, we prove Theorem 1.3 and 1.4. In Section 4, we demonstrate the duality theorem in explicit examples and compute basic invariants. In Section 5, 6, 7, we show various applications and consequences of the main theorems. In Section 8, we study monodromy groups for Lefschetz nets and prove Theorem 1.7.

give more details

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2. PRELIMINARIES

In this section, we revisit classical projective duality from the viewpoint of linear algebra and conormal varieties, setting up the main geometric framework for our later results.

2.1. Linear algebra. For vector space $\mathbb{C}^{N+1}$, we denote $(\mathbb{C}^{N+1})^\vee$ its dual space consisting of linear functionals on $\mathbb{C}^{N+1}$. For any linear subspace $V \subseteq \mathbb{C}^{N+1}$, we denote $V^\perp \subseteq (\mathbb{C}^{N+1})^\vee$ be the subspace consisting of linear functionals vanishing on V . Note that $\text{codim}(V^\perp) = \dim(V)$.

Lemma 2.1. *Let $V^\vee$ denote the dual space of V . Then there is a natural isomorphism of vector spaces*

$$V^\vee \cong (\mathbb{C}^{N+1})^\vee / V^\perp.$$

Proof. Consider the exact sequence

$$0 \rightarrow V \rightarrow \mathbb{C}^{N+1} \rightarrow \mathbb{C}^{N+1}/V \rightarrow 0.$$

Dualizing yields

$$0 \leftarrow V^\vee \leftarrow (\mathbb{C}^{N+1})^\vee \leftarrow (\mathbb{C}^{N+1}/V)^\vee \leftarrow 0.$$

The subspace $(\mathbb{C}^{N+1}/V)^\vee$ consists of those linear functionals on $\mathbb{C}^{N+1}$ that vanish on V . Hence, it is canonically identified with $V^\perp$. This proves the claim. $\square$

Passing to projectivization, $\mathbb{P}V \subseteq \mathbb{P}^N$ is a projective subspace, then $\mathbb{P}V^\perp \subseteq (\mathbb{P}^N)^\vee$ and induces the rational projection

$$(4) \quad p : (\mathbb{P}^N)^\vee \dashrightarrow \mathbb{P}((\mathbb{C}^{N+1})^\vee / V^\perp) \cong \mathbb{P}V^\vee.$$

The indeterminacy locus of p is exactly $\mathbb{P}V^\perp$.

Identifying linear functionals with the set of hyperplanes in $\mathbb{P}^N$, the map (4) can be described by sending a hyperplane H of $\mathbb{P}^N$ to a hyperplane in $\mathbb{P}V$ by intersection:

$$H \mapsto H \cap \mathbb{P}V.$$

It is well-defined as long as H is transversal to (hence it does not contain) $\mathbb{P}V$.

2.2. Projective duality. Now let $X \subseteq \mathbb{P}^N$ be an irreducible subvariety. Let $x \in X^{sm}$ be a closed point of X contained in the smooth locus X^{sm} , then the tangent space $T_x X$ at x has the same dimension as X . We may regard $T_x X$ as a projective subspace of $\mathbb{P}^N$ of dimension $\dim(X)$. We use this convention throughout the paper. Using our notation, $(T_x X)^\perp \subseteq (\mathbb{P}^N)^\vee$ is the conormal space and has dimension $N - \dim(X) - 1$.

The *conormal variety* $N_{\mathbb{P}^N}^\perp X$ of X is defined to be the closure of the union of conormal spaces at smooth points [HK85]

$$(5) \quad N_{\mathbb{P}^N}^\perp X = \overline{\{(x, H) \in X^{sm} \times (\mathbb{P}^N)^\vee \mid T_x X \subseteq H\}}.$$

It records pairs of smooth points and their tangent hyperplanes. It is irreducible and has dimension $N - 1$. Note that here, to require the hyperplane H to contain $T_x X$ is equivalent to finding a hyperplane H tangent to X at x . In the case when X is a hypersurface $f = 0$ of degree at least two, and x is a smooth point, such H is unique and coincides with the tangent hyperplane $T_x X$, which is determined by the normal vector ∇f_x .

Definition 2.2. The *dual variety* of X , denoted $X^\perp$ throughout the paper, is defined to be the image of the projection map $N_{\mathbb{P}^N}^\perp X \rightarrow (\mathbb{P}^N)^\vee$.

Remark 2.3.

Clarify the notational differences between this paper and other papers on projective duality.

There is an incidence correspondence between X and $X^\perp$ shown below.

$$\begin{array}{ccc} & N_{\mathbb{P}^N}^\perp X & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ X & & X^\perp \subseteq (\mathbb{P}^N)^\vee. \end{array}$$

Hence, the conormal space $(T_{x_0} X)^\perp$ is precisely the fiber $\pi_1^{-1}(x_0)$, and is identified with $\pi_2(\pi_1^{-1}(x_0))$, a linear subvariety of the dual variety $X^\vee$. We observe that

Lemma 2.4. *The dimension of the dual variety has a lower bound*

$$\dim(X^\perp) \geq N - 1 - \dim(X).$$

Proof. This is because the dual variety $X^\perp$ contains the linear subspace $(T_{x_0} X)^\perp$, which has dimension $N - 1 - \dim(X)$. $\square$

Remark 2.5. Note that when X is smooth, one has another bound $\dim(X^\perp) \geq \dim(X)$. This follows from Zak's Theorem on Tangencies. See [FL81, Corollary 7.4].

Classical projective duality is a reflexivity property:

Theorem 2.6 (classical projective duality). *The dual variety of $X^\perp$ is isomorphic to X , i.e., there is an isomorphism*

$$X^{\perp\perp} \cong X.$$

Equivalently, projective duality says that for a pair $(x, \alpha) \in X^{sm} \times (X^\perp)^{sm}$, then

$$(6) \quad T_x X \subseteq \alpha^\perp \iff T_\alpha(X^\perp) \subseteq x^\perp,$$

where $\alpha^\perp$ (resp. $x^\perp$) is the hyperplane in $\mathbb{P}^N$ (resp. $(\mathbb{P}^N)^\vee$).

The proof is to show there is an isomorphism of conormal varieties $N_{\mathbb{P}^N}^\perp X \cong N_{(\mathbb{P}^N)^\vee}^\perp X^\perp$ by switching the two coordinates. In fact, the proof is deeply related to symplectic geometry: The product of projective space $\mathbb{P}^N \times (\mathbb{P}^N)^\vee$ lifts to product of vector spaces

$$(7) \quad \mathbb{C}^{N+1} \times (\mathbb{C}^{N+1})^\vee,$$

which is identified with the cotangent bundle of $\mathbb{C}^{N+1}$ and is equipped with a standard symplectic form ω . The lift of the conormal variety $N_{\mathbb{P}^N}^\perp X$ (and $N_{(\mathbb{P}^N)^\vee}^\perp X^\perp$) becomes a conical Lagrangian subvariety (here conical means dilation invariance along fibers). Symplectic geometry guarantees that all the conical Lagrangian subvarieties of (7) arise from conormal varieties and therefore shows the isomorphism $N_{\mathbb{P}^N}^\perp X \cong N_{(\mathbb{P}^N)^\vee}^\perp X^\perp$. We recommend the readers to read [Tev05, Section 1.3], [GKZ08] for details of this perspective.

Example 2.7. Let X be a smooth hypersurface of $\mathbb{P}^N$ of degree $d \geq 2$, then its dual variety is also a hypersurface (in $(\mathbb{P}^N)^\vee$) with degree [EH16, Proposition 2.9]

$$(8) \quad \deg(X^\perp) = d(d-1)^{N-1}.$$

Projective duality implies that if a pair $(x, H) \in X^{sm} \times (X^\perp)^{sm}$ is contained in the smooth locus of the product, then H is tangent to X at x iff x (as a hyperplane in $(\mathbb{P}^N)^\vee$) is tangent to $X^\perp$ at H .

Remark 2.8 (Singularities). When $d \geq 3$, the dual hypersurface $X^\perp$ is singular in codimension one, and in particular non-normal. As an example, when X is a smooth cubic curve in $\mathbb{P}^2$, its dual curve is a sextic curve with 9 cusps.

Remark 2.9 (Defect). For "most" smooth projective varieties X , the dual variety is a hypersurface. Indeed, this happens iff there exists a hyperplane H such that $H \cap X$ has an ordinary node.

In general, for a smooth projective variety X , the *defect* is defined to be

$$\text{def}(X) := N - 1 - \dim(X^\perp),$$

which measures how far the dual variety is from a hypersurface. Varieties with positive defects are often called *dual defective varieties*.

For example, in dimensions one and two, dual defective varieties are linear. The classification is done up to dimension 10 (cf. [Ein86, Ein85, LS87, BFS92]). We will also study defect from the point of view of generic projection in Section 5.

Remark 2.10 (Char p). In positive characteristic, Theorem 2.6 is not valid in general. Those varieties that satisfy Theorem 2.6 are called *reflexive varieties*. Many theorems on dual varieties in char 0 still hold for reflexive varieties in char p . One refers to [Kle86, HK85] for details.

2.3. Plücker formula. Let $C \subseteq \mathbb{P}^2$ be a planar curve of degree d with only ordinary nodes and cusps, then the degree $d^\perp$ and genus $g^\perp$ of its dual curve $C^\perp$ satisfy Plücker relations (cf. [GH94, p.280])

$$(9) \quad d^\perp = d(d-1) - 2\delta - 3\kappa,$$

$$(10) \quad g^\perp = \frac{(d-1)(d-2)}{2} - \delta - \kappa.$$

try to find another reference

Moreover, if the dual curve $C^\perp$ has only ordinary nodes and cusps with numbers $\delta^\perp$ and $\kappa^\perp$, then the curve C has exactly same number of bitangents and flex lines, respectively. The degree d and genus g of C satisfy the dual version of (9) and (10) by switching the invariants of C and $C^\perp$.

When C has worse singularities, there is a generalized Plücker formula [Dol12, 1.51] that replace the invariants of nodes and cusps by Milnor numbers and multiplicities of the singularities.

3. MAIN THEOREMS

In this section, we will prove Theorems 1.3 and 1.4. For the rest of the paper, $X \subseteq \mathbb{P}^N$ will be an irreducible projective variety over complex numbers.

3.1. The setup. We discuss the setup of the theorem. Throughout this section, we use a different convention from the introduction. We take a linear section of X and a generic project of the dual variety $X^\perp$ to a lower-dimensional subspace. (Note that by classical projective duality, we can switch the roles of X and $X^\perp$.)

Let $\mathbb{P}V$ be a subspace of $\mathbb{P}^N$. We consider the subspace section $X \cap \mathbb{P}V$ and denote it as X_V . We denote $X_V^\perp$ the dual variety with respect to the embedding $X_V \subseteq \mathbb{P}V$. We denote $I_{\mathbb{P}}^N$ to be the "0-th order" incidental subvariety

$$I_{\mathbb{P}^N} := \{(x, H) \in \mathbb{P}^N \times (\mathbb{P}^N)^\vee : x \in H\}.$$

Similar for $I_{\mathbb{P}V}$. The conormal varieties $N_{\mathbb{P}^N}^\perp X$ and $N_{\mathbb{P}V}^\perp(X_V)$ are defined by the closure of unions of conormal spaces at smooth points of X and X_V respectively (5). They can be regarded as first-order incidence variety and are naturally included in $I_{\mathbb{P}^N}$ and $I_{\mathbb{P}V}$.

Consider diagram 1. Here, p is the projection from the center $\mathbb{P}V^\perp$. The target is identified to $(\mathbb{P}V)^\vee$ according to Lemma 2.1. The map π is the restriction of p to the dual variety $X^\perp$.

Now we introduce the discriminant locus. It will generalize Definition 1.1 to the singular setting.

Definition 3.1 (Critical and Discriminant locus). Let $\pi : \mathcal{Y} \rightarrow B$ be a flat family of projective varieties. Define the *critical locus* $\mathcal{C} = \mathcal{C}(\pi) \subseteq \mathcal{Y}$ of π to be the closure of the set of points $y \in \mathcal{Y}^{sm}$ such that $(d\pi)_y : T_y \mathcal{Y} \rightarrow T_{\pi(y)} B$ does not have maximal rank, and the *discriminant locus* $\text{disc}(\pi) \subseteq B$ of π to be the closure of $\pi(\mathcal{C})$, where $\mathcal{Y}^{sm}$ is the smooth part of $\mathcal{Y}$.

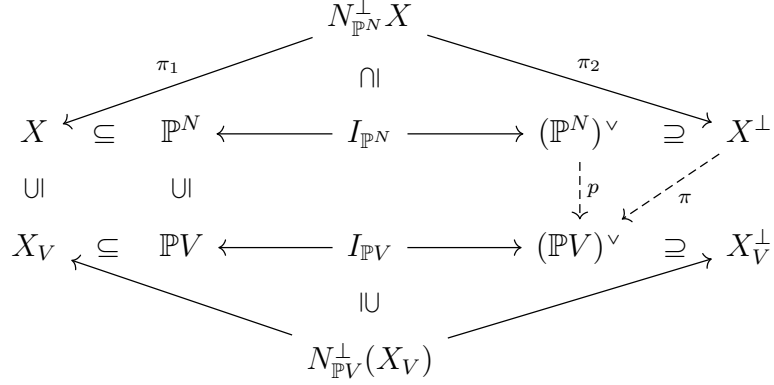


FIGURE 1. Linear section of X and generic projection of $X^\perp$

For the generic projection $\pi : X \dashrightarrow \mathbb{P}^k$, we can extend it to a morphism by blowing up the base locus $\tilde{\pi} : \tilde{X} = \text{Bl}_B X \rightarrow \mathbb{P}^k$. The discriminant locus of π is referred to as the discriminant of $\tilde{\pi}$.

Based on the setup described above, Theorem 1.3 is equivalent to the following:

Theorem 3.2 (cf. Theorem 1.3). *For general choice of linear subspace $\mathbb{P}V \subseteq \mathbb{P}^N$, the discriminant locus of $\pi : X^\perp \dashrightarrow (\mathbb{P}V)^\vee$ is isomorphic to the dual variety $X_V^\perp$ of the plane section X_V of X . In other words, there is an isomorphism*

$$\text{disc}(\pi)^\perp \cong X_V^\perp.$$

To prove this result, we need to introduce a subvariety of the dual variety $X^\perp$ that captures the information of the dual correspondence of X around the subspace $\mathbb{P}V$.

3.2. The critical locus.

Definition 3.3. We define the subvariety $(X|_V)^\perp$ of $X^\perp$ to be

$$(X|_V)^\perp := \pi_2(\pi_1^{-1}(X_V)).$$

In other words, $(X|_V)^\perp$ is the closure of the union of conormal spaces of X at points in $\mathbb{P}V$.

Since for $x \in X_V$, a hyperplane $H \subseteq \mathbb{P}^N$ containing the tangent space $T_x X$ also contains $T_x(X_V)$, we have the following commutative diagram (11):

$$(11) \quad \begin{array}{ccc} X^\perp & \longleftarrow & (X|_V)^\perp \\ \downarrow \pi & & \downarrow \pi|_{(X|_V)^\perp} \\ (\mathbb{P}V)^\vee & \longleftarrow & X_V^\perp. \end{array}$$

The horizontal maps are inclusions, the vertical ones are rational maps restricted from p . To describe the map $\pi|_{(X|_V)^\perp}$, we choose a general point $x \in X_V$ such that x is a smooth point for both X_V and X , then by the description of the map (4), its restriction $\pi|_{(X|_V)^\perp}$ to conormal space $(T_x X)^\perp$ is literally a linear map between two conormal spaces at x :

$$\begin{array}{ccccc}
(12) & (T_x X)^\perp & \xlongequal{\quad} & \{H \in (\mathbb{P}^N)^\vee \mid x \in H \text{ and } T_x X \subseteq H\} & \ni & H \\
& \downarrow \pi|_{(T_x X)^\perp} & & \downarrow & & \downarrow \\
& (T_x X_V)^\perp & \xlongequal{\quad} & \{H_V \in \mathbb{P}V^\vee \mid x \in H_V \text{ and } T_x(X_V) \subseteq H_V\} & \ni & H \cap \mathbb{P}V.
\end{array}$$

Then by further restricting the map $\pi|_{(X|_V)^\perp}$ to an Zariski open subspace, it is the union of all the linear maps (12) when x runs in a Zariski open dense subset of X_V :

$$(13) \quad \pi|_{(X|_V)^\perp} : \bigcup_{x \in X_V^{sm}} (T_x X)^\perp \rightarrow \bigcup_{x \in X_V^{sm}} (T_x X_V)^\perp.$$

We first observe that

Proposition 3.4. *The map $\pi|_{(X|_V)^\perp}$ from (11) is dominant.*

Proof. A general point in $X_V^\perp$ is a hyperplane H_V of $\mathbb{P}V$ tangent to X_V at some point x in the smooth locus, then we note

$$(14) \quad s : H_V \mapsto \text{Span}(H_V, T_x X)$$

defines a rational section of (13). In other words, the above defines a hyperplane H of $\mathbb{P}^N$ and contains $T_x X$, and satisfies $H \cap \mathbb{P}V = H_V$. Hence, the claim follows. $\square$

Now we are close to showing Theorem 3.2. The key argument is to identify the critical locus of π .

Proposition 3.5. *The critical locus $\mathcal{C}(\pi)$ of π is $(X|_V)^\perp$.*

In other words, we will show $(X|_V)^\perp$ is the (closure of) locus where the tangent map $T\pi$ fails to be of the maximal rank. So the Definition 3.3 coincides with [Shi06, Definition 2.15].

By choosing V in general position, we can assume $(X|_V)^\perp$ is not contained in the singular locus of the dual variety, hence its general point $\alpha \in (X|_V)^\perp$ is a smooth point of $X^\perp$, and by projective duality, the conormal space of $X^\perp$ at α , i.e., the set of hyperplanes containing the tangent space $T_\alpha X^\perp$ corresponds to the points $x \in X$ such that the conormal space $(T_x X)^\perp$ contains α .

Proof of Proposition 3.5. First we choose a point $\alpha \in (X|_V)^\perp$ such that α is contained in the smooth locus $(X^\perp)^{sm}$ of the dual variety. Let L_α be the subspace spanned by α and $\mathbb{P}V^\perp$. Equivalently, L_α is the preimage $p^{-1}(\pi(\alpha))$.

Then we note that the point in the critical locus of π is equivalent to

- The tangent map $T_\alpha \pi : T_\alpha(X^\perp) \rightarrow T_{p(\alpha)}\mathbb{P}V^\vee$ fails to be of maximal rank.
- $\iff$ • L_α intersects $X^\perp$ non-transversely at α .
- $\iff$ • The span of $T_\alpha(L_\alpha)$ and $T_\alpha(X^\perp)$ is a proper subspace of $T_\alpha(\mathbb{P}^N)^\vee$.

To show the last statement, we notice that by classical projective duality, the point x corresponds to a hyperplane $x^\perp$ in $(\mathbb{P}^N)^\perp$ containing $T_\alpha(X^\perp)$. On

the other hand, we note $(L_\alpha)^\perp$ is a hyperplane in $\mathbb{P}V$ containing $T_x(X_V)$, and $x \in (L_\alpha)^\perp$. Hence by taking orthogonal complement, we obtain $x^\perp \supseteq L_\alpha$. Thus by identifying $T_\alpha L_\alpha$ with L_α , we conclude that the subspaces $T_\alpha(X^\perp)$ and $T_\alpha L_\alpha$ are collinear, therefore fail to span the whole tangent space. Hence we showed the critical locus of $\pi|_{(X|_V)^\perp}$ is contained in $(X|_V)^\perp$.

Conversely, we choose a general point α in the critical locus of $\pi|_{(X|_V)^\perp}$, then we can assume α is a smooth point of $X^\perp$. Then by the equivalent conditions as above, we conclude that

$$\text{Span}(T_\alpha(X^\perp), T_\alpha(L_\alpha)) \subseteq W,$$

where W is a hyperplane in the dual space $(\mathbb{P}^N)^\vee$. Hence by projective duality (6), $W = x^\perp$ corresponds to a smooth point $x \in X^{sm}$ such that $T_x X \subseteq \alpha^\perp$. On the other hand, the orthogonal complement $(L_\alpha)^\perp$ becomes a hyperplane in $\mathbb{P}V$ and contains the point x . Therefore $x \in X_V$, and $\alpha \in (X|_V)^\perp$. So we showed the other side of inclusion. $\square$

Now we are ready to prove the first main theorem.

Theorem 3.6 (Theorem 3.2). *The discriminant locus of $\pi : X^\perp \dashrightarrow (\mathbb{P}V)^\vee$ is isomorphic to $X_V^\perp$.*

Proof. By Proposition 3.4 and 3.5, the critical locus of π dominates the dual variety $X_V^\perp$, hence $X_V^\perp$ agrees with the critical values in a Zariski open dense subspace. Therefore, by taking closure, the discriminant locus is precisely the dual variety $X_V^\perp$. $\square$

Remark 3.7 (Generality of choice of $\mathbb{P}V$). In Theorem 3.2, the linear subspace $\mathbb{P}V$ needs to be in general position. More specifically, we need the following two conditions:

- (i) $X^{sm} \cap \mathbb{P}V \neq \emptyset$;
- (ii) $(X|_V)^\perp \cap (X^\perp)^{sm} \neq \emptyset$, i.e., the critical locus is not entirely contained in the singular locus of $X^\perp$.

It follows from the Bertini theorem [Jou83] that a general choice of $\mathbb{P}V$ will satisfy the two conditions.

3.3. Characterization of discriminant locus. From now on, we switch back to the convention of Theorem 1.3, i.e., we take generic projection of X and linear section of $X^\perp$.

As a direct consequence of Theorem 1.3, we characterize the discriminant locus D inductively as a discriminant locus of projection to a general hyperplane:

Lemma 3.8. *Let $X \subseteq \mathbb{P}^N$ be an irreducible projective variety. Given a generic projection $\pi : \mathbb{P}^N \dashrightarrow \mathbb{P}^k$, we factor it as a composite of projections to the hyperplanes, as shown in the first row of the following commutative diagram:*

$$(15) \quad \begin{array}{ccccccc} \mathbb{P}^N & \dashrightarrow & \mathbb{P}^{N-1} & \dashrightarrow & \dots & \dashrightarrow & \mathbb{P}^{k+1} & \dashrightarrow & \mathbb{P}^k \\ \text{UI} & \nearrow \pi_N & \text{UI} & \nearrow \pi_{N-1} & & & \text{UI} & \nearrow \pi_{k+1} & \text{UI} \\ X & & D_{N-1} & & \dots & & D_{k+1} & & D_k. \end{array}$$

Let $D_N = X$, we inductively define $D_l \subseteq \mathbb{P}^l$ to be the discriminant locus of the generic projection of $\pi_{l+1} : D_{l+1} \dashrightarrow \mathbb{P}^l$. Then the discriminant locus D of $\pi|_X : X \dashrightarrow \mathbb{P}^k$ is isomorphic to D_k .

Proof. We note that the sequence of generic projections

$$\mathbb{P}^N \dashrightarrow \mathbb{P}^{N-1} \dashrightarrow \dots \dashrightarrow \mathbb{P}^{k+1} \dashrightarrow \mathbb{P}^k$$

corresponds to hyperplanes $H_1, \dots, H_{N-k}$ in $\mathbb{P}^N$, such that for each $1 \leq l \leq N-k$, there is a canonical identification

$$(\mathbb{P}^{N-l})^\vee = H_1 \cap \dots \cap H_l.$$

Then apply Theorem 3.2 inductively to each $1 \leq l \leq N-k$, we get isomorphisms

$$D_{N-l}^\perp = X^\perp \cap (H_1 \cap \dots \cap H_l).$$

In particular, when $i = N-k$, we prove the claim. $\square$

As a consequence, we characterize the discriminant locus D as a generic projection of a hypersurface.

Corollary 3.9. *Let $X \subseteq \mathbb{P}^N$ be a n -dimensional irreducible projective variety. Then any generic projection can be factored as*

$$X \hookrightarrow \mathbb{P}^{n+1} \dashrightarrow \mathbb{P}^k,$$

where the first map f is a generic immersion onto a hypersurface X' of $\mathbb{P}^{n+1}$, and a generic projection g of X' , whose discriminant is precisely D .

Proof. f is obtained by composing the first $N-n-1$ maps. Since the df does not drop rank on the tangent space at a general point, then the critical locus of π is birational to the critical locus of $g : X' \dashrightarrow \mathbb{P}^k$. Therefore, the claim follows. $\square$

One refers to 4.5 for an explicit example.

3.4. Purity of discriminant locus. Now we assume that the generic projection is dominant. We shall prove Theorem 1.4 in this section. We first strengthen Proposition 3.4:

Proposition 3.10. *The map $\pi|_{(X|_V)^\perp}$ is generically one-to-one. In particular, the critical locus is birational onto the discriminant locus.*

Proof. By Corollary 3.9, this reduces to the case where $X^\perp$ is a hypersurface. Then it has a unique tangent hyperplane at a general point, therefore, the rational section (14) defines the unique lifting of $\pi|_{(X|_V)^\perp}$ at a general point, so it is generically one-to-one. $\square$

Now we are ready to prove the second main theorem.

Theorem 3.11 (cf. Theorem 1.4). *Assume that the generic projection*

$$X \dashrightarrow \mathbb{P}^k$$

is dominant (equivalently, $\dim(X) \geq k$). Then the discriminant locus D is either

- *an irreducible hypersurface, if $\text{codim}(X^\perp) < k$, or*
- *union of hyperplanes, if $\text{codim}(X^\perp) = k$, or*
- *empty, if $\text{codim}(X^\perp) > k$.*

here it also make sense to talk about "purity" of critical locus

Proof. If $\text{codim}(X^\perp) > k$, the $X^\perp$ does not intersect with a general linear space of dimension k , hence by Theorem 1.3, the discriminant locus is empty; If $\text{codim}(X^\perp) = k$, the intersection $X^\perp$ with a general $\mathbb{P}^k$ consists of finitely many points, whose dual variety is union of hyperplanes; If $\text{codim}(X^\perp) < k$, then Bertini theorem implies $\mathbb{P}^k \cap X^\perp$ is irreducible, hence by Theorem 1.3, its dual variety, the discriminant D is irreducible as well.

It remains to show D has codimension one. Since $n := \dim(X) \geq k$, we can factor the generic projection as a sequence of generic projections to hyperplanes, and we have a sequence of generic projections of the discriminant loci:

$$X \xrightarrow{\pi_{n+1}} \mathbb{P}^n \supseteq D_n \xrightarrow{\pi_n} \mathbb{P}^{n-1} \supseteq D_{n-1} \rightarrow \cdots \supseteq D_{k+1} \xrightarrow{\pi_{k+1}} \mathbb{P}^k,$$

where each D_l is the discriminant locus of $D_{l+1} \rightarrow \mathbb{P}^l$. We inductively apply Zariski–Nagata theorem, each π_{l+1} is a finite branch cover, hence $D_l \subseteq \mathbb{P}^l$ is a hypersurface. Finally, by Lemma 3.8, D_k is isomorphic to the discriminant locus D , so the result follows. $\square$

Remark 3.12. Alternatively, one can show D has codimension one using Proposition 3.10 and the fact that the critical locus is defined by the 0-th Fitting ideal and has expected codimension one (cf. [Kle77]).

Remark 3.13 (Non-generic projection center). Note that there are choices of $\mathbb{P}V$ satisfying the above conditions, but $\pi|_{(X|_V)^\perp}$ can fail to be generically one-to-one.

As an example, take Q being the smooth quadric surface in $\mathbb{P}^2$, take a tangent hyperplane H such that $H \cap Q$ being union of two lines. Then the discriminant of the corresponding projection of the dual variety

$$p : Q^\perp \cong Q \dashrightarrow \mathbb{P}^2$$

has discriminant being two points $\{p_1, p_2\}$. This is because the projection center c is on the surface, and the map is birational and is everywhere one-to-one except over $\{p_1, p_2\}$ where p contracts the two lines through c .

However, Theorem 3.2 still holds in this case. So the "generality" of the linear subsystem in Theorem 3.2 does not coincide with the "generality" condition needed in Theorem 1.4.

Remark 3.14.

This also implies “purity” of critical locus, i.e. has expected dimension or dimension 1.

4. EXAMPLES

In this section, we illustrate the main theorems in three examples. Most of the results in this section are well known and classical. We derive them from the perspective of projective duality and Plücker formulas.

In Section 4.1, we study generic projection of curves and compute invariants of their dual varieties; In Section 4.2, we take successive generic projections of Veronese surface and find the corresponding discriminant dual are special semistable cubic hypersurfaces related by linear sections; In Section 4.3, we found a net of quadric surfaces produces an elliptic threefold with large Mordell–Weil rank.

4.1. Curves and their duals.

Example 4.1 (Plane curves). Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d , then by Riemann-Roch theorem, a generic projection

$$C \rightarrow \mathbb{P}^1$$

has $d(d-1)$ branch points. On the other hand, the dual curve $C^\perp$ has degree $d(d-1) = |C^\perp \cap (\mathbb{P}^1)^\vee|$ as well (see (8)). By Theorem 1.3, these two sets of $d(d-1)$ points on $\mathbb{P}^1$ are projective equivalent.

Example 4.2 (Spatial Curves). Let $C \subseteq \mathbb{P}^N$ with $N \geq 3$ be a smooth projective curve of degree d and genus g . A generic projection of C into $\mathbb{P}^3$ induces an isomorphism, and the image remains smooth. If we further project it to $\mathbb{P}^2$, then its image $\pi(C)$ is a nodal curve, which is the discriminant locus of a generic projection $C \rightarrow \mathbb{P}^2$. The Plücker formula (9) implies the number of nodes of $\pi(C)$ is

$$\delta = \frac{(d-2)(d-1)}{2} - g.$$

Then by Theorem 1.3, the dual curve of $\pi(C)$ is a plane section of the dual variety $C^\perp$. This allows us to characterize the singularities of the dual variety $C^\perp$ in codimension one (also see [DL81, p.6, Example 1.2]):

Proposition 4.3. *A general plane section $\mathbb{P}^2 \cap C^\perp$ of the dual variety of $C \subseteq \mathbb{P}^N$ is the dual curve of the plane curve $\pi(C)$ and has degree*

$$d^* = 2(d-1) + 2g.$$

Suppose the curve $\mathbb{P}^2 \cap C^\perp$ only has nodes and cusps as its singularities, then the number of nodes δ^ and cusps κ^* equal*

$$\delta^* = 2(d-2)(d-3) + 2g(2d+g-7), \quad \kappa^* = 3(d+2g-2).$$

Proof. The first part follows from Theorem 1.3. Since $\pi(C)$ only have nodes, Plücker formula (9) computes the degree of dual curve as claimed, which is precisely the degree of the dual variety $d^* := \deg(C^\perp) = \deg(\mathbb{P}^2 \cap C^\perp)$.

The assumption on the singularity of $\mathbb{P}^2 \cap C^\perp$ allow one to use the dual version of the Plücker formulas (9) and (10). With d, g , and d^* known, one can solve for δ^* and κ^* . Direct computation leads to the claimed formulas. $\square$

Example 4.4 (Rational normal curve [Zar36]). As a special case, let $C_d \subseteq \mathbb{P}^d$ be the rational normal curve of degree d , then its dual variety is the discriminant hypersurface of degree d binary degree d forms

$$f(x, y) = a_0 x^d + a_1 x^{d-1} y + \cdots + a_d y^d.$$

In other words, the dual variety $C_d^\perp$ consists of $[a_0 : a_1 : \cdots : a_d] \in \mathbb{P}^d$ where $f(x, y)$ has a multiple root. Equivalently, use the identification $\mathbb{P}^n \cong \text{Sym}^n(\mathbb{P}^1)$, the dual variety $C_d^\perp$ parameterizes the set of unordered n points on $\mathbb{P}^1$ with at least two points colliding.

By Proposition 4.3, a general plane section of the dual variety $C_d^\perp$ is a plane curve B of degree $2d-2$ with $2(d-2)(d-3)$ nodes and $3(d-2)$ cusps. This curve is called a *maximal cuspidal rational curve* [Zar36]. Zariski showed that the fundamental group of the complement $\pi_1(\mathbb{P}^2 \setminus B) \cong \pi_1((\mathbb{P}^d)^\vee \setminus C_d^\perp)$ is the braid group

B_d . For the discriminant associated with a higher genus curve, this investigation is generalized by Dolgachev and Ligober in [DL81, Section 3] using Abel-Jacobi maps.

4.2. Veronese surface and sections of cubics. In this section, we provide an example to illustrate Lemma 3.8 that relates generic projections to hyperplanes to successive hyperplane sections of the dual variety.

Example 4.5 (Veronese surface). Let $v_2(\mathbb{P}^2) \subseteq \mathbb{P}^5$ be the Veronese image $v_2 : \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$. Then its dual variety $v_2(\mathbb{P}^2)^\perp$ is the discriminant hypersurface parameterizing singular planar conics [Dol12, Sec. 2.1.1]. Moreover, $M_4 := v_2(\mathbb{P}^2)^\perp$ is a cubic fourfold singular along the Veronese surface $v_2(\mathbb{P}^2)$.

A general hyperplane section $M_3 := M_4 \cap H$ is a cubic threefold singular along the curve $C_4 = v_2(\mathbb{P}^2) \cap H$, which is the rational normal curve of degree four. By Theorem 1.3, M_3 is the dual variety of the generic projection of $v_2(\mathbb{P}^2)$ to $\mathbb{P}^4$, which is actually a smooth embedding due to the fact that the Veronese surface has defective secant variety. In fact, M_4 is the secant variety of $v_2(\mathbb{P}^2)$. M_3 is the secant variety of C_4 . These follow from the properties of the *Severi varieties*, and $v_2(\mathbb{P}^2)$ is one of them (cf. [Zak93, Theorem 2.4]).

If we take one more general hyperplane section M_2 of M_3 , then we get a cubic surface with $4A_1$ singularities. The dual variety of M_2 is the image of a generic projection of $v_2(\mathbb{P}^2)$ to $\mathbb{P}^3$, the *Steiner surface*, a quartic surface singular along three lines. A general plane section of M_2 is a smooth cubic curve, whose dual curve, which by Theorem 1.3 is also the discriminant of the generic projection of $v_2(\mathbb{P}^2) \rightarrow \mathbb{P}^2$, is a planar sextic curve with 9 cusps. Finally, the generic projection of $v_2(\mathbb{P}^2) \rightarrow \mathbb{P}^1$ is a Lefschetz fibration with three singular fibers.

We summarize these projective dualities in the following table:

Generic projection of $v_2(\mathbb{P}^2)$	Hyperplane sections of $M_4 := v_2(\mathbb{P}^2)^\perp$
$v_2(\mathbb{P}^2) \hookrightarrow \mathbb{P}^5$ Veronese surface	$M_4 =$ Cubic fourfold singular along $v_2(\mathbb{P}^2)$
$v_2(\mathbb{P}^2) \hookrightarrow \mathbb{P}^4$ smooth image	$M_3 =$ Cubic threefold singular along C_4
$v_2(\mathbb{P}^2) \hookrightarrow \mathbb{P}^3$ Steiner quartic surface	$M_2 =$ Cubic surface with $4A_1$ singularities
$v_2(\mathbb{P}^2) \xrightarrow{4:1} \mathbb{P}^2$ branched along a sextic curve	$M_1 =$ smooth cubic curve
$v_2(\mathbb{P}^2) \dashrightarrow \mathbb{P}^1$ quartic fibration	$M_0 = 3$ points

TABLE 1. Projective duality between discriminant locus of generic projections of the Veronese surface $v_2(\mathbb{P}^2)$ and successive hyperplane sections of dual variety $v_2(\mathbb{P}^2)^\perp$. The right-hand entries are projective duals of the corresponding varieties on the left. Below the thick line, the projections are dominant.

4.3. An elliptic threefold with large Mordell–Weil rank.

Example 4.6 (Net of quadrics). Let $X = v_2(\mathbb{P}^3)$ be the image of the second Veronese embedding of $\mathbb{P}^3$, and $k = 2$. Then the hyperplane sections family (1) is a general net of quadric surfaces, whose discriminant is a general plane section C of the dual variety $v_2(\mathbb{P}^3)^\perp$. By classical invariant theory [Dol12, p.16], this

dual variety equivalently parameterizes degenerate quadrics on $\mathbb{P}^3$. Therefore

$$(16) \quad v_2(\mathbb{P}^3)^\perp \cong \{A \in \text{Mat}_{4 \times 4} \mid A^T = A, \det(A) = 0\}.$$

is the space of 4×4 symmetric matrices that are degenerate. In particular, $v_2(\mathbb{P}^3)^\perp$ is a normal quartic hypersurface. Hence C is a smooth plane quartic curve, and by classical algebraic geometry, it has 28 bitangents and 24 flex lines.

The second family (2) is a generic projection $v_2(\mathbb{P}^3) \dashrightarrow \mathbb{P}^2$ whose general fiber is a genus one curve as a complete intersection of two quadrics. Theorem 1.3 indicates that

Proposition 4.7. *The discriminant D of the generic projection*

$$(17) \quad v_2(\mathbb{P}^3) \dashrightarrow \mathbb{P}^2$$

is an irreducible plane curve of degree 12, with 28 nodes and 24 cusps.

Proof. It suffices to show that D is general in the moduli and hence has only nodes and cusps. This is true because the space of quartic curves arising from plane sections of the quartic hypersurface (16) dominates the moduli of quartic curves [Ter10]. Then the claim follows from the Plücker formulas. $\square$

The base locus of (17) is the complete intersection of three general quadrics and consists of 8 points, which is called a Cayley octad. Passing to the blowup of the base locus produces an elliptic threefold

$$\tilde{\pi} : \widetilde{v_2(\mathbb{P}^3)} \rightarrow \mathbb{P}^2,$$

smooth over the complement of D , and has

- nodal fibers over smooth points of D ;
- 28 Kodaira type I_2 fibers (union of two conics meeting at two points) over the nodes of D ;
- 24 cuspidal fibers over the cusps of D .

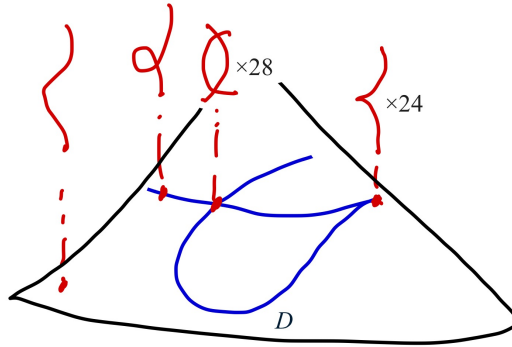


FIGURE 2. Discriminant curve of generic projection of the Veronese threefold

There is an interest in constructing an elliptic threefold with a large Mordell-Weil rank over $\mathbb{P}^2$ (or rational surfaces). The current record is 10 [GW22], but it is isotrivial. Other approaches include [CAL14, HK11]. We provide a non-isotrivial example below.

Proposition 4.8. *The above construction produces an elliptic threefold over $\mathbb{P}^2$ which is non-isotrivial, and has Mordell-Weil rank 7. (cf. Example 4.6)*

Proof. Blowup of the Veronese image of $\mathbb{P}^3$ at base locus of generic projection, which consists of 8 points. So we arise at an elliptic threefold $v_2(\mathbb{P}^3)$ with Picard rank 9. The discriminant locus is irreducible, and a general fiber over it is nodal curve (see Example 4.6), so by Shioda-Tate-Wazir formula [Waz04], we have

$$\rho(\widetilde{v_2(\mathbb{P}^3)}) = 1 + \rho(\mathbb{P}^2) + r + 0.$$

Therefore, the Mordell-Weil rank is $r = 7$ as claimed. $\square$

This was also obtained in [Tot08, Theorem 7.2]. One also refers to [PS18] for elliptic 3-folds arising from a special net of quadrics.

5. SMOOTH FIBRATION OF DUAL DEFECTIVE VARIETIES

In this section, we further investigate Theorem 1.4 in the case that the discriminant locus D is empty. The following will prove Corollary 1.5.

Corollary 5.1 (Smooth fibration). *Let $X \subseteq \mathbb{P}^N$ be a projective variety such that $\mathbb{P}V \cap X^\perp = \emptyset$, and let B be the base locus of the projection $\pi : X \dashrightarrow \mathbb{P}V^\vee$. Then the induced morphism*

$$\tilde{\pi} : \mathrm{Bl}_B X \rightarrow \mathbb{P}V^\vee$$

is a submersion over the smooth locus X^{sm} of X . In particular, if X is smooth, then $\tilde{\pi}$ is a smooth proper morphism.

Proof. According to the proof of Theorem 3.2, $\mathbb{P}V \cap X^\perp$ being empty implies that the critical locus of $(\mathrm{Bl}_B X)^{\mathrm{sm}} \rightarrow \mathbb{P}V^\vee$ is empty. Hence if X is smooth, so is $\mathrm{Bl}_B X$, and the morphism $\tilde{\pi}$ is smooth. $\square$

As an application, we recover a classical result that dual defect does not occur in $\dim(X) = 1$ and 2 [Ein86, Proposition 3.1].

Proposition 5.2 (Dual defect for curve and surface). *Let $X \subseteq \mathbb{P}^N$ be a smooth projective variety of dimension 1 or 2. Suppose X is not a linearly embedded line or plane, then the dual variety $X^\perp$ is a hypersurface.*

Proof. Let $X = C$ be a curve. If $C^\perp$ is not a hypersurface, by Corollary 5.1, a generic projection $C \rightarrow \mathbb{P}^1$ is unramified, which forces $C \cong \mathbb{P}^1$ to be a linearly embedded line.

Let $X = S$ be a surface. If $S^\perp$ is not a hypersurface, by Corollary 5.1, by blowing up the base locus (which consists of $\deg(S)$ distinct points) of a general pencil $\tilde{S} \rightarrow S$, there is a smooth family

$$\pi : \tilde{S} \rightarrow \mathbb{P}^1.$$

If fiber has genus zero, then $\tilde{S}$ is a Hirzebruch surface. Otherwise, π must be isotrivial, since for $g \geq 1$, the moduli space M_g of smooth genus g curves does not contain a (complete) rational curve. Let F be a general fiber. Since the family is smooth and $\mathbb{P}^1$ has trivial fundamental group, and $\mathrm{Aut}(F)$ is finite for a $g(F) \geq 2$, π is in fact a trivial family, hence $\tilde{S} \cong F \times \mathbb{P}^1$. (For $g(F) = 1$

case, there is a section endowing the group structure on the generic fiber. The monodromy preserves the group structure, so can be globally trivialized.)

Hence $\tilde{S}$ is either a Hirzebruch surface or $F \times \mathbb{P}^1$ for a curve F with $g(F) \geq 1$. They are both minimal surfaces, contradicting the fact that $\tilde{S}$ arises as a nontrivial blowup of S , except that $\tilde{S}$ is the first Hirzebruch surface $\mathbb{F}_1$, which arises from blowing up a linearly embedded $\mathbb{P}^2$ at one point. Hence, the claim follows. $\square$

Example 5.3 (Defective 3-fold dual). The first example of the defective dual variety (of a smooth projective variety) starts with dimension three: the Segre 3-fold

$$X = \mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$$

has dual defect one. By Corollary 5.1, by blowing up a general $\mathbb{P}^3$ -section B of X , which is a twisted cubic, there a smooth morphism

$$\mathrm{Bl}_B X \rightarrow \mathbb{P}^1,$$

and every fiber is isomorphic to the first Hirzebruch surface $\mathbb{F}_1$.

Remark 5.4 (Singular total space). Note that when X is singular, the condition in Corollary 5.1 does *not* guarantee that every fiber the morphism

$$\mathrm{Bl}_B X \rightarrow \mathbb{P}^k$$

is smooth, even passing to an appropriate desingularization. In fact, this family is isotrivial and every fiber is singular, but there can be fibers with worse singularities. This is because the singular strata of X may not be smooth over the base, and this is not captured by the classical projective duality. See an example below.

Example 5.5 (dual of twisted cubic). We take X to be the dual variety of a twisted cubic $C \cong X^\perp \subseteq \mathbb{P}^3$. Since a general $\mathbb{P}^1$ will miss the curve C , i.e., $C \cap \mathbb{P}^1 = \emptyset$, Theorem 1.3 implies that the generic projection $X \dashrightarrow \mathbb{P}^1$ has empty discriminant. However, the morphism by blowing up the base locus B

$$\tilde{\pi} : \mathrm{Bl}_B X \rightarrow \mathbb{P}^1$$

is not smooth, even after passing to the minimal desingularization. In fact, X is a quartic surface and has transversal cuspidal singularities along the osculating curve C' . The curve C' is isomorphic to the twisted cubic C itself. Hence B consists of four points and the general fiber of $\tilde{\pi}$ is a quartic curve with three cusps. The family $\tilde{\pi}$ is isotrivial, but has four reducible fibers corresponding to the four branch points of $C' \rightarrow \mathbb{P}^1$. The normalization $\widetilde{\mathrm{Bl}_B X}$ of $\mathrm{Bl}_B X$ is a weak del Pezzo surface of degree four, and the composition map

$$\widetilde{\mathrm{Bl}_B X} \rightarrow \mathrm{Bl}_B X \rightarrow \mathbb{P}^1$$

has general fiber $\mathbb{P}^1$, but contains 4 reducible fibers.

5.1. Reducible discriminant. We have seen many examples where the discriminant locus of a generic projection is irreducible (cf. Section 4). In this section, we focus on reducible discriminant, corresponding to case (ii) of Theorem 1.4.

First of all, we note that for *smooth* projective varieties X with dual defect r , a generic projection $X \dashrightarrow \mathbb{P}^{r+1}$ is dominant and has discriminant being the union of hyperplanes. However, by Remark 2.9 and Proposition 5.2, nontrivial examples are rare and start from dimension three. Here, we provide an example when X is singular in dimension two.

Example 5.6 (dual of spatial curves). Let $C \subseteq \mathbb{P}^3$ be a projective curve of degree d , and let $X = C^\perp$ be the dual variety (so $d = 3$ is in Example 5.5). Then X is a surface in the dual space $(\mathbb{P}^3)^\vee$. It is singular along a curve C' , called the osculating curve associated with C . Choosing a general plane $\mathbb{P}^2 \subseteq \mathbb{P}^3$, then the projection

$$X \rightarrow (\mathbb{P}^2)^\vee$$

is a branched cover, and its discriminant consists of d lines.

To see this, by Theorem 1.3, the discriminant is dual of $\mathbb{P}^2 \cap C = d$ points, i.e., the set of d lines. From another point of view, this can be seen from the fact that the dual variety $C^\perp$ is also the tangential variety of the osculating curve C' [Dol12, p.640], so if one point lies in the discriminant locus, the whole tangent line also lies in the discriminant locus.

The converse also holds, in other words, if the generic projection of a surface $X \subseteq \mathbb{P}^3$ to $\mathbb{P}^2$ has discriminant being the union of lines, then the dual variety of X is a curve.

5.2. Defect and hyperplane section. In this section, we discuss the dual defect for X and its general hyperplane section. The following is shown for the smooth case in [Tev05, Theorem 5.3]. Here we generalize to the singular case. Here we define the defect for (singular) projective variety X in the same way as in Remark 2.9.

Lemma 5.7. *Let $X \subseteq \mathbb{P}^N$ be an irreducible projective variety. Let $r = \text{def}(X)$ be the defect of X , i.e., $r = N - 1 - \dim X^\perp$. Let H be a general hyperplane, then*

$$\text{def}(X \cap H) = \begin{cases} r - 1, & \text{if } r \geq 1, \\ 0, & \text{if } r = 0. \end{cases}$$

Proof. When $r \geq 1$, then by dimension reason, for a general hyperplane H , the discriminant locus of $\pi : X^\perp \dashrightarrow H^\vee$ is the image $\pi(X^\perp)$, which has the same dimension as $X^\perp$, so the codimension decreases by one. By Theorem 1.3, $\pi(X^\perp)$ is the dual variety of $X \cap H$, hence by definition the defect decreases by one, i.e., $\text{def}(X \cap H) = \text{def}(X) - 1$. Otherwise, if $r = 0$, then the projection to $H^\vee$ is a branch cover. It follows from the Zariski–Nagata theorem that the discriminant locus is still a hypersurface, hence $X \cap H$ has no defect. $\square$

6. DEGREE OF POLAR LOCUS

Recall that $N_{\mathbb{P}^N}^\perp(X)$ is the conormal variety (5). It has dimension $N - 1$ and admits two projections: the first projection π_1 to X is generically a projective bundle, and the second projection π_2 has image the dual variety $X^\perp$. Following

[Kle86, Cha. II], for each $0 \leq i \leq N$, one defines the intersection number

$$(18) \quad r_i(X) = \int_{N_{\mathbb{P}^N}^\perp(X)} \pi_1^* H^i \cdot \pi_2^* (H')^{N-1-i},$$

where $H = c_1 \mathcal{O}_{\mathbb{P}^N}(1)$ and $H' = c_1 \mathcal{O}_{(\mathbb{P}^N)^\vee}(1)$ are hyperplane classes on $\mathbb{P}^N$ and the dual space $(\mathbb{P}^N)^\vee$, respectively.

These intersection numbers have special geometric meanings. For example, we assume X is irreducible and denote $n = \dim(X)$ as usual, then note that by the projection formula, $r_n = \deg(X)$ is the degree of X ; $r_i = 0$ when $i > n$; and

$$r_0 = \begin{cases} \deg(X^\perp), & \text{if } X^\perp \text{ is a hypersurface;} \\ 0, & \text{otherwise.} \end{cases}.$$

In general, r_i counts the number of $N - 1 - i$ planes that are tangent to X in a general linear pencil tangent to X .

Definition 6.1. [[Kle86, p. 186]] The i -th polar locus of X with respect to a general codimension $i + 2$ linear subspace $A_i \subseteq \mathbb{P}^N$ is defined to be

$$X(A_i) = \overline{\{p \in X^{sm} \mid \dim T_p X \cap A_i = n - i - 1\}},$$

consisting of points where the corresponding tangent plane intersects A_i having larger dimension than expected.

The intersection number (18) has the following equivalent characterization:

Proposition 6.2. [Kle86, p.187] For $0 \leq i \leq n$, the intersection number r_i coincides with the degree of i -th polar locus

$$r_i = \deg(X(A_i)).$$

In general, one can talk about the class of i -th polar locus $[X(A_i)]$ in the Chow group. There is an enumerative geometry formula to compute this class [Kle77, IV,21].

6.1. Relation to generic projection. We will give an equivalent characterization from the point of view of generic projection. Note that the subspace $A_i \subseteq \mathbb{P}^N$ induces a rational map $\mathbb{P}^N \dashrightarrow \mathbb{P}^{i+1}$, which restricts to X is a generic projection

$$\pi : X \dashrightarrow \mathbb{P}^{i+1},$$

as long as A_i is in general position.

We observe that

Lemma 6.3. The polar locus $X(A_i)$ coincides with the critical locus of the projection π .

Proof. Note that the projection $\mathbb{P}^N \dashrightarrow \mathbb{P}^{i+1}$ coincides with the dual of the inclusion $A_i^\perp \subseteq (\mathbb{P}^N)^\vee$. So by Definition 3.3 (and switch X and its dual), the critical locus of π is $\pi_1(\pi_2^{-1}(A_i^\perp \cap X^\perp))$. The general point of this set consists of $x \in X$, such that there exists of a hyperplane H_α satisfying $T_x X \subseteq H_\alpha$ and $A_i \subseteq H_\alpha$. Hence $T_x X$ and A_i are collinear and $\dim(T_x X \cap A_i) \geq n - i - 1$, so x is contained in the i -th polar locus $X(A_i)$, and vice versa. $\square$

Recall that we have showed that via the generic projection π , the critical locus is generically one-to-one to the discriminant locus (cf. Proposition 3.10), Lemma 6.3 implies that

Corollary 6.4. *When $0 \leq i \leq n$, the intersection number r_i (18) coincides with the degree of the discriminant locus of the generic projection $\pi : X \dashrightarrow \mathbb{P}^{i+1}$.*

6.2. Recovering classical results. As a direct consequence of the above, we can easily recover various classical results on the degree of the polar locus.

Theorem 6.5. [HK85] $r_i \neq 0$ iff $\text{def}(X) \leq i \leq n$, where $\text{def}(X)$ is the defect of $X^\perp$ (cf. Definition 2.9).

Proof. Follow from Corollary 6.4, $r_i \neq 0$ iff the discriminant of $\pi : X \dashrightarrow \mathbb{P}^{i+1}$ has nontrivial discriminant, which by Theorem 1.4, is equivalent to $\text{def}(X) \leq i$. $\square$

Theorem 6.6. [Pie78] *If H is a general hyperplane, then $r_i(X) = r_{i-1}(X \cap H)$.*

Proof. Let H be the hyperplane corresponding to a general hyperplane $\mathbb{P}^i \subseteq \mathbb{P}^{i+1}$, the by restriction the generic projection $\pi : X \dashrightarrow \mathbb{P}^{i+1}$ to H , we have $\pi|_H : X \cap H \dashrightarrow \mathbb{P}^i$, whose discriminant locus $\text{disc}(\pi|_H)$ is precisely the intersection $\text{disc}(\pi) \cap \mathbb{P}^i$. $\square$

By induction on taking hyperplane sections, one obtain

Theorem 6.7 (Segre, 1912). [Kle86, p.190] *There is an equality*

$$r_i(X) = r_0(X_i),$$

where X_i is a section of X by a general codimension i linear subspace.

6.3. Degree of discriminant curve. Now we assume that $X \subseteq \mathbb{P}^N$ is smooth and non-defective, then the discriminant locus of generic projection

$$X \dashrightarrow \mathbb{P}^2$$

is an irreducible curve D , which has degree equal to the degree of the first polar class [Kle77, IV,31]. Explicitly,

$$(19) \quad \deg(D) = r_1(X) = \deg(c_1(\Omega_X)) + (n+1) \deg(X).$$

One refers to an example below.

Example 6.8 (Surfaces in $\mathbb{P}^3$). Let $S \subseteq \mathbb{P}^3$ be a smooth surface of degree d , then its dual variety $S^\perp$ has degree (see (8))

$$\deg(S^\perp) = d(d-1)^2.$$

- Project from a general line, the general fiber of

$$\pi : S \dashrightarrow \mathbb{P}^1$$

is a degree d plane curve and has exactly $d(d-1)^2$ nodal fibers. This is precisely the Lefschetz fibration as we mentioned in the introduction.

- Project from a general point, then we obtain a degree d branched cover

$$(20) \quad \pi : S \rightarrow \mathbb{P}^2$$

whose branch locus is an irreducible curve D . By Theorem 3.2, its dual curve is a plane section of the dual variety $S^\perp \cap (\mathbb{P}^2)^\vee$. Using this fact, we can determine the geometry of D :

This formula is wrong. How to fix it?

Proposition 6.9. *The branch curve D of (20) has degree $d(d-1)$, number of nodes and cusps equal to $\delta = \frac{1}{2}d(d-1)(d-2)(d-3)$, and $\kappa = d(d-1)(d-2)$, respectively.*

As an example, when $d = 2$, the branch curve is a smooth conic; when $d = 3$, the branch curve is a plane sextic with 6 cusps¹.

Proof. By the adjunction formula, $K_S = \pi^*K_{\mathbb{P}^2} + \tilde{D}$, where $\tilde{D} \subseteq S$ is the ramification divisor. Use the fact that $K_S = \mathcal{O}_S(d-4)$ and $\pi^*K_{\mathbb{P}^2} = \mathcal{O}_S(-3)$, we find $\tilde{D} = \mathcal{O}_S(d-1)$. Hence

$$(21) \quad \deg(D) = \deg(\tilde{D}) = \deg(\mathcal{O}_S(d-1)) = d(d-1).$$

This proves the first part.

Now, by [CF11], the ramification locus $\tilde{D}$ is smooth, hence by evaluating the degree of the adjunction formula on $\tilde{D}$, we find its genus

$$(22) \quad g(\tilde{D}) = \frac{1}{2}d(d-1)(2d-5) + 1.$$

Theorem 3.2 implies that the dual plane curve $D^\perp$ is a plane section of the dual variety $S^\perp \cap (\mathbb{P}^2)^\vee$. Therefore, by (8), we obtain

$$\deg(D^\perp) = \deg(S^\perp) = d(d-1)^2.$$

On the other hand, by [CF11], the discriminant curve D only have nodes and cusps, hence their numbers are computed by the Plücker formulas (9), (10). $\square$

7. SINGULARITIES OF DISCRIMINANT CURVE (SUFFICIENTLY AMPLENESS)

In this section, we show that for projections $\pi: X \rightarrow \mathbb{P}^2$ dual to a general plane section $\mathbb{P}^2 \subseteq (\mathbb{P}^N)^\vee$, if X is sufficiently ample, singular fibers of π must be of type A_1 , $A_1 + A_1$ or A_2 .

First of all, when an embedding $X \subseteq \mathbb{P}^N$ is sufficiently ample (in fact, 2-jet ample is enough, see [Shi08]), then the dual variety of X is a hypersurface. So for a general plane section $X^\perp \cap \mathbb{P}^2$ of dual variety is an irreducible curve. Simpson characterized the singularities of this curve.

Theorem 7.1. [Sim93, Proposition 6.1] *Let X be a smooth projective variety, and the embedding $X \hookrightarrow \mathbb{P}^N$ is sufficiently ample. Then a general plane section of the dual variety $X^\perp \cap \mathbb{P}^2$ is an irreducible plane curve with only nodes and cusps. Moreover, let $s \in \mathbb{P}^2 \cap X^\perp$ be a point on the discriminant curve, then s is*

- a smooth point iff X_s has a single ordinary double point (type A_1);
- a node iff X_s has exactly two ordinary double points (type $A_1 + A_1$);
- a cusp iff X_s has exactly one ordinary cusp (type A_2).

Here in Simpson's proof, it requires more ampleness. An embedding $X \hookrightarrow \mathbb{P}^N$ being sufficiently ample is requires for any pairs of points $(y, z) \in X$, the natural restriction map

$$H^0(X, \mathcal{O}_X(1)) \rightarrow \mathcal{O}_X(1)/\mathfrak{m}_y^4 \oplus \mathcal{O}_X(1)/\mathfrak{m}_z^4$$

¹In fact, one can show the 6 cusps lie on a conic, so these sextic curves lie in a distinguished component of Zariski pair [Zar29].

These invariants are special case in Prop 2.6 of [CMT01]. Maybe we can compute for all hypersurfaces and then say in the 2-dimensional case, it agree with the reference

can delete part of the proof

is surjective. For example, a line bundle $\mathcal{L} = \mathcal{O}_X(1)$ which is 7-jet ample will suffice (see [BS93, p.358]).

Theorem 7.2 (Dolgachev-Libgober, Roberts, etc). *Under the assumption that the curve $X^\perp \cap \mathbb{P}^2$ has only nodes and cusps, the number equals*

$$\delta =, \kappa = .$$

Our main theorem of this section can be stated as follows:

Theorem 7.3. *Suppose X is a smooth projective variety, and $X \hookrightarrow \mathbb{P}^N$ is a sufficiently ample embedding. Then a generic projection*

$$X \dashrightarrow \mathbb{P}^2$$

has discriminant locus D being an irreducible curve with only nodes and cusps as its singularities. Fibers $\pi^{-1}(x)$ over smooth points $x \in D$ have a single double point, over nodes have exactly two double points, and over cusps have exactly one singularity of type A_2 . Moreover, the numbers of nodes and cusps of D are equal to

$$\delta =, \kappa = .$$

Remark: There are two ways to compute the number. One use projective duality and Theorem 7.1; The other use the Euler characteristic (in [CMT01], authors used this method).

Remark: In the case of surfaces, these numbers are computed in [CMT01, CF11].

Let $X \subseteq \mathbb{P}^N$ (with $N \geq 3$) be an irreducible projective variety with the dual variety $X^\perp$ being a hypersurface, then the generic projection to a plane

$$X \dashrightarrow \mathbb{P}^2$$

has discriminant locus D being is an irreducible plane curve (if X is not a point). We want to understand the singularities of D .

By Theorem 1.3, D is the dual curve of plane section of $X^\perp$. The degree of $X^\perp$ is known (cf. [DL81, (2,1)])

$$d^\perp = \sum_{i=0}^n (i+1) \deg(c_{r-i}(\Omega_X^1) \cdot H^i),$$

where $n = \dim(X)$ and H is the hyperplane class. Assume moreover that $X^\perp$ has "expected" type of singularities (nodes and cusps) in codimension one. Then there are also formulas for the number of cusps κ^* and nodes δ^* for the general plane section $X^\perp \cap (\mathbb{P}^2)^\vee$ [DL81, (2,2),(2.3)]:

$$\kappa^\perp = \kappa^\perp(n, c_i(\Omega_X^1), \deg(X)),$$

$$\delta^\perp = \delta^\perp(n, c_i(\Omega_X^1), c_j(N_X \mathbb{P}^N), \deg(X), \kappa),$$

depending on n , degree of X , the Chern classes of cotangent bundle $c_i(\Omega_X^1)$ and the normal bundle $c_j(N_X \mathbb{P}^N)$.

Then use the Plücker relations, we obtain

Proposition 7.4. *For a given smooth projective variety $X \subseteq \mathbb{P}^N$, and $X \dashrightarrow \mathbb{P}^2$ a generic projection, with discriminant curve D . Assume that $X^\perp$ is a hypersurface with only nodes and cusps in codimension one. Then there are formulas*

computing the degree, the number of nodes and cusps of D depending on the dimension, the degree of X , the Chern classes of the normal bundle $N_X \mathbb{P}^N$, and the Chern classes of the cotangent bundle Ω_X^1 .

Proof. By the discussion above, d^*, δ^*, κ^* are known. Then we can compute the degree d and genus g of D using Plücker relations (9) and (10).

Since the (geometric) genus of D and $D^\perp$ coincides, we can use the same formulas (9) and (10), with the role of D and $D^\perp$ reversed and obtain the number of nodes and cusps of the discriminant curve D . $\square$

Remark 7.5. The assumption that $X^\perp$ is a hypersurface is the most interesting case for studying Lefschetz net (or generic projection to $\mathbb{P}^2$). Otherwise, plane sections of $X^\perp$ is either finitely many points, or empty, which correspond to the second and the third cases of Theorem 1.4; The assumption on expected singularities in codimension one can be guaranteed if the embedding $X \subseteq \mathbb{P}^N$ is sufficiently ample (cf. [Shi06], [Sim93, Proposition 6.1]).

When X is a curve or a hypersurface, then very ample $\implies$ sufficiently ample. In general, we don't know how to relax this condition.

One refers to Proposition 6.9 and Example 4.5 for specific examples.

7.1. Proof of Theorem 7.3. The idea of the proof of theorem 7.3 is to set up the relation between fibers of $\pi: X \rightarrow \mathbb{P}^2$ and hyperplane sections $\{X_s\}_{s \in \mathbb{P}^2}$ corresponding to a plane section $\mathbb{P}^2 \subseteq (\mathbb{P}^N)^\vee$ dual to the given generic projection, so that Simpson's theorem applies. Firstly observe that

Lemma 7.6. *Let $X \subseteq \mathbb{P}^N$ be a projective variety, $X^\perp \subseteq (\mathbb{P}^N)^\vee$ be the projective dual of X and $\mathbb{P}^2 \subseteq (\mathbb{P}^N)^\vee$ a plane section of $(\mathbb{P}^N)^\vee$, then there is a one-to-one correspondence*

$$\left\{ \begin{array}{l} \text{fibers of the dual} \\ \text{projection } X \rightarrow \mathbb{P}^2 \end{array} \right\} \xleftrightarrow{1:1} \left\{ \begin{array}{l} \text{1-dimensional} \\ \text{subspaces of } \mathbb{P}^2 \subseteq (\mathbb{P}^N)^\vee \end{array} \right\}$$

between pencils $\mathbb{L}$ in $\mathbb{P}^2$ and centers of the pencil $Z(\mathbb{L}) = \bigcap_{H \in \mathbb{L}} X_H$ where $X_H = X \cap H$ is the corresponding hyperplane section.

Proof. Pick $s \in \mathbb{P}^2$, we can consider the canonical projection $\mathbb{P}^2 \xrightarrow{p_s} \mathbb{P}^1$. Preimage of any $[t] \in \mathbb{P}^1$ under the composition $\pi_s := p_s \circ \pi$ is a hyperplane section of X containing X_s . The composition π_s then defines a pencil $\mathbb{L}_s$ on X with center $\pi^{-1}(s)$. $\square$

Fix $H \in \mathbb{P}^2$, the space of centers of pencils containing H can be described as the image of the projection

$$p_{H^\perp}: \mathbb{P}^2 \setminus \{H^\perp\} \rightarrow \mathbb{P}^1$$

where any choice $[K^\perp] \in \mathbb{P}^1$ determines a pencil $p_{H^\perp}^{-1}([K^\perp])$ and hence the center $Z_{H,K} := X_H \cap K = H \cap K \cap X$.

Note that the map $p_{H^\perp}$ is the restriction of the projection $p_{H^\perp}: (\mathbb{P}^N)^\vee \setminus \{H^\perp\} \rightarrow H^\vee$ to the dual variety of H , so that $\mathbb{P}^1$ provides a pencil on X_H of the space of all possible centers, where intersections of $\mathbb{P}^1$ with $X_H^\perp \subseteq H^\vee$ correspond to centers having singularities.

Based on the dicussions, we can make the following construction: let

$$\mathcal{F} := \{(\mathbb{P}^2, H^\perp, [K^\perp]) \mid \mathbb{P}^2 \subseteq (\mathbb{P}^N)^\vee, H^\perp \in \mathbb{P}^2, [K^\perp] \in \mathbb{P}^1\}$$

be the partial flag variety and

$$\mathcal{P} := \{(H^\perp, K_H^\vee) \mid H^\perp \in (\mathbb{P}^N)^\vee, K_H^\vee \in H^\vee\}$$

be the universal bundle of pencils in hyperplanes of $(\mathbb{P}^N)^\vee$. Each $X \subseteq \mathbb{P}^N$ induces a subvariety

$$\mathcal{P}_X := \overline{\{(H^\perp, K_H^\vee) \mid K_H^\vee \in X_H^\perp, H^\perp \in (\mathbb{P}^N)^\vee\}}$$

of closure of dual varieties of $X_H = H \cap X$ in H . The subvariety $\mathcal{P}_X$ induces a stratification $\mathcal{S}_X$ of $\mathcal{P}$ where the top stratum is $\mathcal{P} \setminus \mathcal{P}_X$, and the codimension k stratum is given by the smooth locus of the subvariety $(\mathcal{P}_X)^{(k-1)}$ where $(\mathcal{P}_X)^{(0)} = \mathcal{P}_X$. Note that there is a natural forgetful map

$$\tilde{\pi}: \mathcal{F} \rightarrow \mathcal{P}: (\mathbb{P}^2, H^\perp, [K^\perp]) \mapsto (H^\perp, K_H^\vee)$$

which is surjective with fiber $\text{Gr}(2, N)$.

Definition 7.7. *The triple $(\mathbb{P}^2, H^\perp, [K^\perp])$ is general if $\tilde{\pi}(\mathbb{P}^2, H^\perp, [K^\perp])$ does not lie in $\mathcal{P}_X$.*

Note that genericity of the triple is equivalent to smoothness of the center $Z_{H,K}$ determined by this triple. Therefore it suffices for us to show the following Proposition:

Proposition 7.8. *Let $X \hookrightarrow \mathbb{P}^N$ be a projective variety and let $S: \mathbb{P}^2 \hookrightarrow (\mathbb{P}^N)^\vee$ be a plane section. Write*

$$\mathcal{F}_S := \left\{ (S, H^\perp, [K^\perp]) \mid H^\perp \in S(\mathbb{P}^2), [K^\perp] \in H^\vee \right\}$$

to be the fiber of $\mathcal{F}$ at S , then for general choices of S , $\tilde{\pi}(\mathcal{F}_S)$ only intersects with strata of $\mathcal{S}_X$ of codimension at most 2.

Proof. Note that the intersection depth depends only on the singularity type of the center $Z_{H,K}$, and for fixed pair (H, K) , replacing H or K by any pairs (H, L) or (L, K) where $L = aH + (1-a)K$ does not change the space $Z_{H,K}$, and hence all such elements (S, H, L) or (S, L, K) belong to the same stratum. Therefore $\tilde{\pi}(\mathcal{F}_S)$ can be constructed from a $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$ as follows: let $\mathbb{P}^1 \subseteq S$ be any one-dimensional subspace and $\mathbb{P}^1 \times_S \mathcal{F}_S$ the restricted $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$. We can then consider the space

$$\mathcal{H} := \left\{ (H^\perp, [K^\perp]) \mid \begin{array}{l} \exists H_0^\perp, H_1^\perp \in S \text{ \& } a \in \mathbb{C} \\ H^\perp = aH_0^\perp + (1-a)H_1^\perp, \\ [K^\perp] \in H^\vee \end{array} \right\},$$

which can be shown to be isomorphic to $\tilde{\pi}(\mathcal{F}_S)$. In particular, it suffices to consider the intersection depth of the embedding $P \hookrightarrow \mathcal{P}$ with $\mathcal{P}_X$ where P is a general $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$, and the Proposition follows from the genericity of the choice of P . $\square$

In order to apply Simpson's result, we need one more Lemma about jet ampleness:

Lemma 7.9. *Let $X \subseteq \mathbb{P}^N$ be a k -jet very ample projective subvariety for some k , then any subvariety $Y \subseteq X$ is k -jet very ample.*

Proof. Pick closed points $x_1, \dots, x_\ell \in Y$ and $k_1, \dots, k_\ell$ such that $\sum_{i=1}^\ell k_i = k + 1$, it suffices for us to show that the restriction map

$$H^0(Y, \mathcal{O}_Y(1)) \rightarrow \bigoplus_{i=1}^\ell \mathcal{O}_Y(1) / \mathfrak{m}_{Y, x_i}^{k_i} \mathcal{O}_Y(1)$$

is surjective. Note that X is k -jet very ample, so that the restriction map

$$H^0(X, \mathcal{O}_X(1)) \rightarrow \bigoplus_{i=1}^\ell \mathcal{O}_X(1) / \mathfrak{m}_{X, x_i}^{k_i} \mathcal{O}_X(1)$$

is surjective. The closed immersion induces a commutative diagram

$$\begin{array}{ccc} H^0(X, \mathcal{O}_X(1)) & \twoheadrightarrow & \bigoplus_{i=1}^\ell \mathcal{O}_X(1) / \mathfrak{m}_{X, x_i}^{k_i} \mathcal{O}_X(1) \\ \downarrow & & \downarrow \\ H^0(Y, \mathcal{O}_Y(1)) & \longrightarrow & \bigoplus_{i=1}^\ell \mathcal{O}_Y(1) / \mathfrak{m}_{Y, x_i}^{k_i} \mathcal{O}_Y(1) \end{array}$$

where the vertical arrow $\bigoplus_{i=1}^\ell \mathcal{O}_X(1) / \mathfrak{m}_{X, x_i}^{k_i} \mathcal{O}_X(1) \rightarrow \bigoplus_{i=1}^\ell \mathcal{O}_Y(1) / \mathfrak{m}_{Y, x_i}^{k_i} \mathcal{O}_Y(1)$ is surjective. This proves the desired surjectivity. $\square$

We are now ready to prove theorem 7.3. A general choice of a $\mathbb{P}^1$ -bundle P over $\mathbb{P}^1$ in $\mathcal{P}$ can be separated into two steps: we first choose a general $\mathbb{P}^1$ in $(\mathbb{P}^N)^\vee$ and then choose a general $\mathbb{P}^1$ -fiber on each point of the line. A general $\mathbb{P}^1$ intersects transversely with the dual variety $X^\perp$ of X , which is a hypersurface in $(\mathbb{P}^N)^\vee$, at finitely many points. We can pick general $\mathbb{P}^1$ s over each point, and interpolate generally to get the $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$. Note that a general $\mathbb{P}^1$ does not intersect with $\mathcal{P}_X$, and hence the center defined by these points are smooth. The codimension 1 intersection corresponds to the intersection of a general $\mathbb{P}^1$ fiber over some H with $\mathcal{P}_X|_H$, which is the codimension 1 stratum of $H^\vee$, and hence the corresponding center has a unique nodal singularity.

A general one-parameter family of $\mathbb{P}^1$ intersects with codimension 2 stratum of some fiber $H^\vee$, at finite number of points, which by Simpson's result 7.1 corresponds to centers with $A_1 + A_1$ or A_2 singularities. Finally, for intersections of $\mathbb{P}^1$ over $H^\perp \in X^\perp$ with $X_H^\perp$, the subvariety $H_{x_0}^\vee$ of $H^\vee$ passing through the unique double point $x_0 \in X \cap H$ is a hyperplane of $H^\vee$, where for any general $K_H^\perp \in H_{x_0}^\vee$, the intersection $K \cap H \cap X$ has a unique singular double point x_0 . Therefore for general choice of $\mathbb{P}^1$ over points on $X^\perp$, the family of centers has at worst a unique ordinary double point. $\square$

This also implies that

Claim 7.10. *For a general point $t \in D$, the fiber $\pi^{-1}(t)$ has an ordinary nodal singularity.*

One compares to the hyperplane case due to Lefschetz.

Write down a proof: we can prove it by relating the intersection of $\mathbb{P}^1$ -bundle over $\mathbb{P}^1$ to the discriminant locus. The claim is better to be put in section 7.

8. SURJECTION ONTO THE BRAID GROUP.

Revise this part.

In this section, we show how the algebro-geometric statement Theorem 1.4 leads to structural results on monodromy groups of Lefschetz fibrations. Recall that given any smooth projective variety X , and a projective embedding $X \hookrightarrow \mathbb{P}^N$, a *Lefschetz pencil* on X is a general choice of a projective line in the dual variety $(\mathbb{P}^N)^\vee$, intersecting transversely with the dual variety $X^\perp$.

Equivalently via projective duality, a Lefschetz pencil corresponds to a birational map $X \dashrightarrow \mathbb{P}^1$ with discriminant locus disjoint union of finitely many points on $\mathbb{P}^1$. A choice of a *Lefschetz net* is a general choice of $\mathbb{P}^2$ in $(\mathbb{P}^N)^\vee$ dual to a birational map to $\mathbb{P}^2$ which, after blow-up, is a proper map to $\mathbb{P}^2$ with discriminant locus C dual to $\mathbb{P}^2 \cap X^\perp$, an irreducible curve for general enough choice of $\mathbb{P}^2$.

With an additional general choice of a pencil on $\mathbb{P}^2$ and hence on C , for each $\mathbb{P}^1 \hookrightarrow \mathbb{P}^2$ in this pencil, by pulling-back the fibration to $\mathbb{P}^2$ we obtain a Lefschetz fibration with general fiber Z a codimension-two linear section of X . This family can be viewed as a “pencil of Lefschetz fibrations”, where monodromy action on a pencil $Z \hookrightarrow Y$ is induced from the monodromy action on automorphism groups of $\mathbb{P}^1 \setminus C$, isomorphic to $\mathbb{P}^1$ removing $\deg C$ many closed points.

Proposition 8.1 ([FM12, Section 9.1]). *There is a short exact sequence of groups*

$$1 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \pi_1 [C(\mathbb{P}^1, \deg C)] \rightarrow \text{Mod}(\mathbb{P}^1 \setminus C) \rightarrow 1$$

where $\text{Mod}(\mathbb{P}^1 \setminus C)$ is the group of automorphisms of the open Riemann surface $\mathbb{P}^1 \setminus C$, and $C(\mathbb{P}^1, \deg C)$ is the configuration space of $\deg C$ distinct points on $\mathbb{P}^1$.

The main goal of this subsection is to show that the fundamental group of $\mathcal{O}^{sm}(Y)$, the universal family of hyperplane sections of $Y \hookrightarrow \mathbb{P}^N$, has a surjection onto the mapping class group $\text{Mod}(S_{0,m})$ of $S_{0,m} := \mathbb{P}^1 \setminus \{m \text{ distinct points}\}$. Assume that Y has no defects and is a hyperplane section of some $X \hookrightarrow \mathbb{P}^{N+1}$, which for instance holds for general hypersurfaces of $\mathbb{P}^N$, then a choice $\mathbb{L}$ of a general pencil of hyperplane sections on Y can be regarded as

Proposition 8.2. *Let $Z \hookrightarrow Y \hookrightarrow \mathbb{P}^N$ be smooth projective varieties such that Y is extendable with no defect and Z is a general hyperplane section of Y , and let $\mathbb{L}$ be a pencil on Y with Z as one of the pages, then the fundamental group of universal family of hyperplane sections of Y surjects onto the mapping class group $\text{Mod}(S_{0,m})$ where $S_{0,m}$ is $\mathbb{P}^1$ removing m many distinct points.*

Proof. We consider a smooth projective variety $Y \hookrightarrow X \hookrightarrow \mathbb{P}^{N+1}$ having Y as one of its hyperplane sections, and consider a pencil $\mathbb{L}_1$ on X . The pencils $\mathbb{L}_1$ and $\mathbb{L}$ (lifted to a pencil of X) together give a two-dimensional linear system on X , hence inducing a birational fibration $X \dashrightarrow \mathbb{P}^2$, where the discriminant locus $C \subseteq \mathbb{P}^2$, as we have shown in Theorem 1.4, is an irreducible curve in $\mathbb{P}^2$ of degree m .

Construct the group homomorphism before stating the Proposition, i.e. put some of the proofs before this Proposition.

The pencil $\mathbb{L}_1$ induces a pencil on $(\mathbb{P}^2, C)$, and as X is irreducible, it's classically known ([Voi07, Theorem 3.22]) that the fundamental group of $\mathbb{P}^1$ removing discriminant locus of the map $\mathbb{L}_1^\perp: X \dashrightarrow \mathbb{P}^1$ surjects onto the fundamental group of the universal family of hyperplane sections $\pi_1(\mathcal{O}^{sm}, Y)$. On the other hand, $\mathbb{L}_1^\perp$ factors through the map $(\mathbb{P}^2, C) \dashrightarrow \mathbb{P}^1$ where the discriminant locus of $\mathbb{L}_1^\perp$ completely come from the discriminant locus of $\pi|_C: C \dashrightarrow \mathbb{P}^1$.

As based loops of $\mathbb{P}^1$ removing discriminant locus D_π of π act on the fiber of π at the base point, which is diffeomorphic to $\mathbb{P}^1$ removing $m = \deg C$ many points, it suffices to show that $\pi_1(\mathbb{P}^1 \setminus D_\pi, *)$ acts transitively on $C \cap \mathbb{P}^1$, hence $\pi_1(\mathbb{P}^1 \setminus D_\pi, *)$ admits a surjection onto $\mathfrak{S}_m$ which factors through the forgetful map $\text{Mod}(\mathbb{P}^1 \setminus C) \rightarrow \mathfrak{S}_m$ forgetting “directions” of the Dehn twist. As there're monodromies from $\pi_1(\mathbb{P}^1 \setminus D_\pi, *)$ permuting two of the points in $\mathbb{P}^1 \cap C$ rotating the two points exactly once, the map $\pi_1(\mathbb{P}^1 \setminus D_\pi, *) \rightarrow \text{Mod}(\mathbb{P}^1 \setminus C) \rightarrow \mathfrak{S}_m$ are all surjective.

Finally, $\pi_1(\mathbb{P}^1 \setminus D_\pi, *)$ acting transitively on $C \cap \mathbb{P}^1$ is a Corollary of the fact that when C is irreducible, monodromy actions on fibers of $C \dashrightarrow \mathbb{P}^1$ are transitive [Voi07, Exercise 2 of Chapter 3]. $\square$

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